

Long-term visual outcomes and side effects of ruthenium plaque brachytherapy in 310 eyes with small choroidal melanoma

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► Additional supplemental material is published online only. To view, please visit the journal online (<https://doi.org/10.1136/bjo-2023-324686>).

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Received 6 October 2023
Accepted 9 January 2025
Published Online First
29 January 2025

ABSTRACT

Background/aims To report the long-term visual outcomes and side effects in patients with small choroidal melanoma (CM) undergoing ruthenium-106 (Ru-106) plaque brachytherapy.

Methods Retrospective, interventional, consecutive series of small CM ≤ 2.5 mm in height and ≤ 16 mm in largest basal diameter treated with Ru-106 plaque with a median radiation dose of 100 Gy prescribed to tumour apical height.

Results 310 patients (160 men) with a mean (SD) age of 58.4 ± 14.1 years met the study inclusion criteria. The median follow-up was 57.5 (range 3.4–170.5) months. The mean tumour baseline thickness was 1.9 ± 0.4 (range 0.4–2.5) mm. The mean baseline logarithm of the minimum angle of resolution (logMAR) of best corrected visual acuity (BCVA) was 0.21 ± 0.4 (Snellen equivalent 6/9; range -0.1 to 1.5). Mean final BCVA was ≤ 0.3 logMAR (Snellen equivalent 6/12) in 161 patients (54.2%), better than 1.0 logMAR (Snellen equivalent 6/60) and < 0.3 logMAR in 59 patients (19.9%), and ≥ 1.0 logMAR in 77 patients (25.9%). Kaplan-Meier estimates of poor final VA (≥ 1.0 logMAR) were 13.2% at 5 years and 54.5% at 10 years. Early and late complications developed in 20.6% and 42.6% of cases, respectively. Radiation maculopathy was the most frequent late side effect (29% of patients) with Kaplan-Meier rates of 18.2%, 31.7% and 42.1% at 3, 5 and 10 years, respectively. Older age, lipofuscin, proximity to fovea, final tumour elevation and radiation maculopathy predicted visual loss ≥ 5 Snellen lines.

Conclusion Despite early and late complications in 20.6% and 42.6% of cases, Ru-106 brachytherapy for small CM allows retention of BCVA ≤ 0.3 logMAR in half of the eyes.

INTRODUCTION

Early treatment of small choroidal melanoma (CM) poses challenges in balancing the risk to vision from radiotherapy with the potential risk to life from an untreated or delayed treated melanoma. The strategy of initial observation of small melanocytic lesions with treatment offered only if signs of malignant transformation arise¹ has been abandoned in many centres in favour of early intervention.² Partly, this is from the knowledge that CMs as small as 3 mm in largest basal diameter (LBD)

WHAT IS ALREADY KNOWN ON THIS TOPIC

- ⇒ About half of choroidal melanomas (CMs) treated with iodine-125 have a best-corrected visual acuity (BCVA) worse than 6/60 at 5 years from treatment.
- ⇒ Information on the risks of plaque radiotherapy for small CM is still limited, with only a handful of reports focusing on visual outcomes and radiation side effects after plaque brachytherapy for tumours of this size category.

WHAT THIS STUDY ADDS

- ⇒ In small CM treated with ruthenium-106 (Ru-106) plaque brachytherapy, a BCVA worse than 6/60 was observed at 5 years in 13.2% and at 10 years in 54.5% of eyes. Early and late radiation side effects developed in 20.6% and 42.6% of eyes, respectively, and radiation maculopathy was the most common.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

- ⇒ This study adds important information relative to vision loss, early and late radiation-related complications of small CM treated with Ru-106 plaque, and offers useful guidance to patient counselling.

can metastasise.³ Nonetheless, only 28%–30% of all newly diagnosed uveal melanomas in the USA are small.^{4–6}

First introduced by Moore almost a century ago,⁷ brachytherapy is the mainstay conservative radiation treatment for small-sized and medium-sized CM. The Collaborative Ocular Melanoma Study showed no significant survival advantage for medium-sized CMs treated by iodine-125 (I-125) brachytherapy or enucleation.⁸ Brachytherapy may preserve the globe in the majority of cases but carries morbidity of local radiation-related side effects.^{9–11}

It is known that about half of choroidal tumours treated with I-125 have a best-corrected visual acuity (BCVA) $\leq 6/60$ at 5 years from treatment,¹¹ but the information on the risks of radiation treatment for small tumours is still limited. Herein, the purpose of our study is to add information on the long-term visual outcomes and side effects for CMs having a thickness of < 2.5 mm treated with



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To cite: Gallo B, Hussain R, Al-Jamal R, et al. *Br J Ophthalmol* 2025;109:809–815.

ruthenium-106 (Ru-106) plaque brachytherapy with a radiation dose prescribed to the tumour apical height.

METHODS

This study was conducted in the London Ocular Oncology Service based at Moorfields Eye Hospital and St. Bartholomew's Hospital.

The electronic medical records of all patients with small CM treated with Ru-106 plaque brachytherapy between October 2004 and May 2019 were retrospectively reviewed. Eligible were patients affected by small CM having a height ≤ 2.5 mm and a base ≤ 16 mm measured with ultrasound (US).

Patient data included sex, age, ethnicity, medical history, date of diagnosis and treatment, and tumour laterality. Tumours were retrospectively staged according to the 8th edition of the American Joint Committee on Cancer Tumour, Node, Metastasis System,¹² and location, quadrant, distance to foveola and optic disc, height, LBD, aspect (dome, diffuse, collar-stud), pigmentation and extrascleral extension were recorded. The distance of posterior tumour margin from the optic disc and foveola was determined by reviewing fundus photographs, assuming the optic disc to have a horizontal diameter of 1.5 mm and using this as a reference.

Each visit was a comprehensive examination including Snellen BCVA, intraocular pressure, slit-lamp biomicroscopy, indirect funduscopy, colour or pseudocolour fundus pictures (Topcon, Japan; Optos, UK), fundus autofluorescence (Optos or Spectralis, Heidelberg Engineering, Germany), optical coherence tomography (OCT, Spectralis) and B-scan US (ACUSON S 2000, Siemens Healthcare, Germany). Fluorescein and indocyanine green angiography (Optos or Spectralis) were performed in case of uncertain diagnosis.¹³

Treatment was offered to lesions showing growth on serial USs or fundus pictures, or having ≥ 3 of the following risk factors: thickness ≥ 2 mm, subretinal fluid (SRF), visual symptoms, orange pigment, distance from optic disc ≤ 3 mm, hollow internal US acoustic reflectivity.¹⁴ Suspicious lesions were monitored every 4 months to look for growth or new risk factors.

Patients were treated with Ru-106 plaque brachytherapy (Eckert & Ziegler BEBIG GmbH, Germany), and plaque size (12, 15 or 20 mm) and type (circle or notched) were recorded. Our surgical technique is reported elsewhere.¹⁵ Radiation doses and timing were calculated according to the manufacturer's plaque simulator data. An apex dose of 80–100 Gy giving a scleral dose of 250–350 Gy was prescribed.

Follow-up data included visual outcomes, early (developing in the first 6 months after treatment) and late treatment-related side effects namely radiation maculopathy, non-proliferative and proliferative retinopathy, optic neuropathy, cataract, toxic tumour syndrome (TTS), neovascular glaucoma, scleral necrosis and retinal vascular occlusions. Radiation maculopathy comprised features including macular teleangiectasia, microaneurysms, exudation or ischaemia. Non-proliferative features of radiation retinopathy included retinal oedema, haemorrhages, exudates, cotton wool spots and capillary drop-out, and proliferative features consisted in retinal or disc neovascularisation, preretinal haemorrhage, vitreous haemorrhage and tractional retinal detachment. Radiation optic neuropathy was characterised by optic disc swelling followed by pallor and atrophy. TTS was characterised by vasculopathy and exudative retinal detachment. Radiation-induced cataract was defined by the onset of or asymmetric progression of lens opacity after plaque.

Statistical analysis

Statistical analysis was performed using Microsoft Excel (V.16.41) and Stata software (V.14.1, StataCorp LP). Results are presented as mean $\pm$ SD when normally distributed, or as median (IQR and range) when not. Paired t-test was used to test statistically significant differences between initial and final BCVA and tumour size. Hypothesis testing used two-tailed statistics, and a significance was defined by p value of ≤ 0.05 .

Kaplan-Meier analysis estimated the cumulative probability of radiation-related side effects, final BCVA ≥ 1.0 logMAR and visual acuity loss ≥ 5 Snellen lines. Univariate analyses with Cox proportional hazards model were performed to identify factors predictive of side effects, and subsequent multivariate analyses were performed using the forward stepwise method for factors significant at $p \leq 0.05$ in the univariate analyses. HRs with 95% CIs were calculated. The baseline and final Snellen BCVA were converted to logarithm of the minimum angle of resolution (logMAR). Visual acuities of counting fingers, hand motion, light perception and no light perception were assigned logMAR designations of 1.9, 2.3, 2.7 and 3, respectively. BCVA for enucleated eyes was recorded as no light perception.

RESULTS

From October 2004 to May 2019, three hundred seventy-eight eyes with small CM received Ru-106 plaque radiotherapy. We excluded patients that lacked adequate follow-up (43); received other treatment prior to plaque (19); with choroidal tumours involving ciliary body and/or iris (6). 310 eyes from 310 patients with small CM (298 stage T1a, 2 stage T1c, 10 stage T2a) were included, with a mean age of 58.4 ± 14.1 (range 25–89) years. The median follow-up was 57.5 (range 3.4–170.5) months. 296 patients (95.5%) were white, 160 (51.6%) were men, and the right eye was affected in 163 (52.6%) patients. There were symptoms in most of the patients (194, 62.6%), and the mean duration was 3.6 ± 3.7 (range 0.25–24) months. Systemic comorbidities included 31 (10.3%) patients with diabetes, 105 (34%) with hypercholesterolemia and 127 (41%) with hypertension.

Patient demographics and baseline tumour characteristics are reported in [table 1](#).

Plaque size was 12 mm in 5 eyes (1.6%), 15 mm in 43 eyes (13.9%) and 20 mm in 262 eyes (84.5%); plaque shape was notched for 91 (29.4%) tumours and circular in the remaining cases.

Local tumour control and survival for this cohort have been reported by the authors in a separate study.¹⁵

Visual acuity outcomes

[Table 2](#) summarises the baseline and final BCVA for all eyes. Eyes with a posterior pole (0.26 ± 0.42 logMAR) and juxtapapillary tumours (0.19 ± 0.4 logMAR) had worse baseline BCVA. The difference between baseline and final BCVA was significant for all tumour locations ($p < 0.001$) except peripheral tumours, and SRF absence was associated with better baseline BCVA.

Pre-treatment BCVA was ≤ 0.3 logMAR (6/12 Snellen) in 251 eyes (81%), between 0.3 and 1.0 logMAR (6/12 and 6/60) in 45 eyes (14.5%) and ≥ 1.0 logMAR (6/60) in 14 eyes (4.5%). Final BCVA was ≤ 0.3 logMAR in 161 eyes (54.2%), better than 1.0 logMAR and < 0.3 logMAR in 59 eyes (19.9%) and ≥ 1.0 logMAR in 77 eyes (25.9%). BCVA improvement, defined as an increase of at least 0.3 logMAR, occurred in 22 eyes (7.4%).

Kaplan-Meier estimates of poor final VA (≥ 1.0 logMAR) were 13.2% at 5 years and 54.5% at 10 years, and estimates for VA

Table 1 Baseline demographic data and tumour characteristics of patients with small choroidal melanoma (n=310)

Features	n (%)
Age at diagnosis (years), median (mean, range)	60 (58.4, 25–89)
Ethnicity	
White British	256 (82.6)
White other	40 (12.9)
Asian	1 (0.3)
Unknown	13 (4.2)
Gender	
Male	160 (51.6)
Female	150 (48.4)
Laterality	
Right	163 (52.6)
Left	147 (47.4)
Ocular symptoms at presentation	194 (62.6)
Absent	116 (37.4)
Blurred vision	105 (54.4)
Photopsia	93 (48.2)
Floaters	7 (3.6)
Visual field defect	27 (14)
Metamorphopsia	13 (6.7)
Ocular pain	1 (0.5)
Symptom duration (months), median (mean, range)	3 (3.6, 0.25–24)
Follow-up duration (months), median (mean, range)	57.5 (64.3, 3.4–170.5)
Distance to optic disc (mm), median (mean, range)	3 (3.4, 0–15)
Posterior pole tumours	3 (3±1.8, 0–8)
Juxtapapillary tumours	0 (0.4±0.7, 0–3)
Peripheral tumours	5 (5.9±3.2, 1.5–15)
Posterior pole and peripheral tumours	3 (3.1±1.4, 1–6)
Distance to foveola (mm), median (mean, range)	2.5 (3.1, 0–15)
Posterior pole tumours	0.5 (1.2±1.4, 0–6)
Juxtapapillary tumours	3 (2.5±1.7, 0–6)
Peripheral tumours	6 (6.7±2.9, 3.5–15)
Posterior pole and peripheral tumours	0.5 (0.9±1.1, 0–3)
Location	
Posterior pole, macula	145 (46.8)
Juxtapapillary	61 (19.7)
Nasal	20
Temporal	23
Inferior	7
Superior	11
Periphery	95 (30.6)
Nasal	35
Temporal	24
Inferior	6
Superior	30
Posterior pole and periphery	9 (2.9)
Subretinal fluid	274 (88.4)
Localised	246
Exudative retinal detachment	28
Thickness (mm), median (mean, range)	2 (1.9, 0.4–2.5)
LBD (mm), median (mean, range)	7.1 (7.5, 2.9–16)
2.9≤LBD<6 mm	76 (24.5)
6≤LBD<9 mm	159 (51.3)
9≤LBD<12 mm	60 (19.4)
12≤LBD≤16 mm	15 (4.8)
Shape	

Continued

Table 1 Continued

Features	n (%)
Dome	282 (91)
Collar-stud	3 (1)
Diffuse	25 (8)
TNM stage	
T1a	298 (96.1)
T1b	–
T1c	2 (0.7)
T2a	10 (3.2)
Colour	
Pigmented	235 (75.9)
Amelanotic	25 (8)
Mixed pigmentation	50 (16.1)
Associated features	
Extraocular extension	2 (0.7)
Halo	4 (1.3)
Baseline BCVA	
≤0.3 logMAR	251 (81)
Between 0.3 and 1.0 logMAR	45 (14.5)
≥1.0 logMAR	14 (4.5)

BCVA, best-corrected visual acuity; LBD, largest basal diameter; LogMAR, logarithm of the minimum angle of resolution; TNM, tumour, node, metastasis.

loss ≥5 Snellen lines were 15.1% at 5 years and 59.8% at 10 years (table 3 and figure 1).

Factors predictive of final BCVA ≥1.0 logMAR on univariable and multivariate analyses included older age (HR 1.03, 95% CI 1.01 to 1.05), posterior pole location (HR 2.77, 95% CI 1.64 to 4.65), baseline BCVA (HR 3.34, 95% CI 2.35 to 4.75), lipofuscin (HR 2.57, 95% CI 1.28 to 5.19), closer distance to disc (HR 0.78, 95% CI 0.7 to 0.87) and radiation maculopathy (HR 1.73, 95% CI 1.10 to 2.72) (online supplemental table S1).

Tumour dimensions and response to plaque brachytherapy

Mean pre-treatment tumour height was 1.9±0.4 (range 0.4–2.5) mm, and mean LBD was 7.5±2.2 (range 2.9–16) mm; mean final tumour elevation was 1.2±0.6 (range 0–5.1) mm. The difference between baseline and final tumour thickness was significant (p<0.0001), with mean reduction of 36.8%.

Treatment-related side effects

Early (within 6 months from treatment) side effects developed in 64 (20.6%) eyes, with diplopia most commonly observed (47 patients, 15.2%), requiring surgery or botulinum toxin in two patients with peripheral superior tumours and self-limiting in the remainder of cases. Uveitis was the second most common early side effect developing in 6 (1.9%) eyes and resolving with topical steroids. Online supplemental table S2 summarises the treatment-related side effects.

Late side effects developed in 132 (42.6%) eyes after a mean of 40.7±25.8 (5.1–161.2) months from treatment. The most frequent was radiation maculopathy occurring in 90 (29%) eyes at a mean of 34.9±17.6 (5.1–84.5) months. Radiation retinopathy was observed in 61 (19.7%) eyes and was non-proliferative in 50 (16.1%) eyes and proliferative in 11 (3.5%) eyes. Radiation optic neuropathy was observed

Table 2 Baseline and final best-corrected visual acuity (BCVA) based on tumour location and presence of subretinal fluid (SRF)

	Baseline BCVA (logMAR) mean (median, range)	Final BCVA (logMAR) mean (median, range)	P value
All tumours	0.21±0.4 (0.2, −0.1 to 1.5)	0.5±0.7 (0.2, −0.1 to 3)	<0.001
Posterior pole tumours	0.26±0.42 (0.2, −0.1 to 2.7)	0.7±0.7 (0.5, −0.1 to 3)	<0.0001
Juxtapapillary tumours	0.19±0.4 (0, −0.2 to 1.9)	0.6±0.7 (0.3, −0.2 to 2.3)	0.0015
Peripheral tumours	0.13±0.4 (0, −0.2 to 2.3)	0.19±0.4 (0, −0.1 to 2.3)	0.21
Posterior pole and peripheral tumours	0.14±0.25 (0, −0.1 to 0.6)	0.6±0.6 (0.3, 0.2 to 1.8)	0.019
Tumours with SRF	0.22±0.41 (2, −0.2 to 2.3)	0.61±0.7 (0.3, −0.1 to 3)	<0.001
Tumours without SRF	0.09±0.17 (0, −0.2 to 0.5)	0.23±0.47 (0, −0.2 to 1.5)	0.019

LogMAR, logarithm of the minimum angle of resolution.

in 22 (7.1%) eyes, the majority with juxtapapillary tumours (13, 59.1%). TTS developed in 2 (0.6%) eyes. Radiation-induced cataract was observed in 3 (1%) eyes. There were no cases of neovascular glaucoma and scleral necrosis. **Table 3** and **figure 2** report Kaplan-Meier analysis of late side effects.

Factors predictive of late radiation-related side effects with the highest HRs were for radiation maculopathy SRF (HR 2.8, 95%CI 1 to 7.84) and plaque size (HR 1.31, 95%CI 1.07 to 1.61), for non-proliferative radiation retinopathy plaque size (HR 1.27, 95%CI 1.01 to 1.6) and final tumour elevation (HR 1.93, 95%CI 1.15 to 3.24), for proliferative radiation retinopathy baseline BCVA (HR 3.16, 95%CI 1.15 to 8.69) and for optic neuropathy closer distance to disc (HR 0.6, 95%CI 0.38 to 0.94) (online supplemental table S1).

Management of late side effects

A total of 42 (46.7%) eyes affected by radiation maculopathy confirmed on fluorescein angiography received treatment, consisting of intravitreal anti-vascular endothelial growth factor (VEGF, bevacizumab 1.25 mg/0.05 mL) injections for 40 eyes, macular laser for 1 eye and a combination of both for 1 eye. The injection protocol consisted of a loading phase of three monthly injections followed by treat and extend regimen. The average number of injections was 4.8±5.5 (range 1–32). As expected, a significant difference in final BCVA between treated (mean BCVA 0.56±0.52

logMAR) and untreated (mean BCVA 1±0.79 logMAR) eyes was observed (p=0.004).

Among the 11 patients with proliferative radiation retinopathy, 2 were untreated due to poor visual prognosis and 9 received pan-retinal photocoagulation (6 cases) or vitreo-retinal surgery (3 cases). TTS was managed in one eye with laser transpupillary thermotherapy (TTT).

DISCUSSION

Early treatment of small CM has three main objectives: to achieve local tumour control, to prevent metastatic disease and to preserve useful vision. While for large tumours, the vision impairment is predominantly due to their clinical characteristics, such as retinal detachment, small lesions are often located in proximity of the fovea and optic disc, and vision loss occurs by several mechanisms: from the tumour, its effects on surrounding structures or the side effects of treatment. The desire to treat such tumours early is counter-balanced by the risks to vision from the proposed treatment.

Currently, there are a handful of reports on long-term outcomes and side effects of brachytherapy for small choroidal tumours^{2 16–21} (online supplemental table S3). The variability in methods of radiation dose calculation, radioisotopes and additional treatments prior to plaque² limit objective comparisons. In the current study, we have reported the long-term visual outcomes and rates of side effects of Ru-106 brachytherapy in

Table 3 Kaplan-Meier analysis of outcomes for plaque brachytherapy for small choroidal melanoma in 310 consecutive patients

		Kaplan-Meier estimates, no affected patients/no unaffected patients (%) (95% CI)					
Outcomes	n (%)	1 year	2 years	3 years	5 years	10 years	14 years
<i>Late complications</i>							
Radiation maculopathy	90 (29)	10/290 (3.3) (1.8 to 6)	27/252 (9.2) (6.4 to 13.2)	50/188 (18.2) (14.1 to 23.4)	77/103 (31.7) (26.1 to 38.2)	90/15 (42.1) (35.1 to 49.9)	90/2 (42.1) (35.1 to 49.9)
Radiation retinopathy	61 (19.7)						
Non-proliferative	50 (16.1)	4/296 (1.3) (0.5 to 3.4)	15/261 (5.1) (3.1 to 8.4)	26/203 (9.7) (6.7 to 13.9)	45/118 (19.6) (14.9 to 25.6)	51/16 (25.2) (19.2 to 32.7)	51/2 (25.2) (19.2 to 32.7)
Proliferative	11 (3.5)	0/300 (0) (NA)	0/276 (0) (NA)	4/220 (1.5) (0.6 to 4.1)	9/141 (4.4) (2.3 to 8.3)	10/21 (5.7) (2.9 to 11)	11/1 (24.5) (5.7 to 73.9)
Radiation optic neuropathy	22 (7.1)	1/299 (0.3) (0.1 to 2.3)	8/270 (2.8) (1.4 to 5.5)	13/215 (4.7) (2.8 to 8)	20/139 (8.5) (5.5 to 13)	22/22 (10.7) (6.8 to 16.5)	22/2 (10.7) (6.8 to 16.5)
<i>Visual acuity</i>							
Final BCVA ≥1.0 logMAR*	77 (25.9)	0/287 (0) (NA)	5/265 (1.8) (0.8 to 4.3)	17/212 (6.7) (4.2 to 10.5)	29/141 (13.2) (9.3 to 18.5)	71/22 (54.5) (44.4 to 65.2)	76/2 (86.4) (60.6 to 98.6)
VA loss ≥5 Snellen lines	91 (30.6)	1/287 (0.3)(0.05 to 2.4)	5/265 (1.8) (0.8 to 4.3)	17/212 (6.7) (4.2 to 10.6)	33/141 (15.1) (10.9 to 20.7)	80/22 (59.8) (49.9 to 70)	90/2 (92.9) (76.5 to 99.2)
*Enucleated eyes excluded. BCVA, best-corrected visual acuity; NA, not available; VA, visual acuity.							

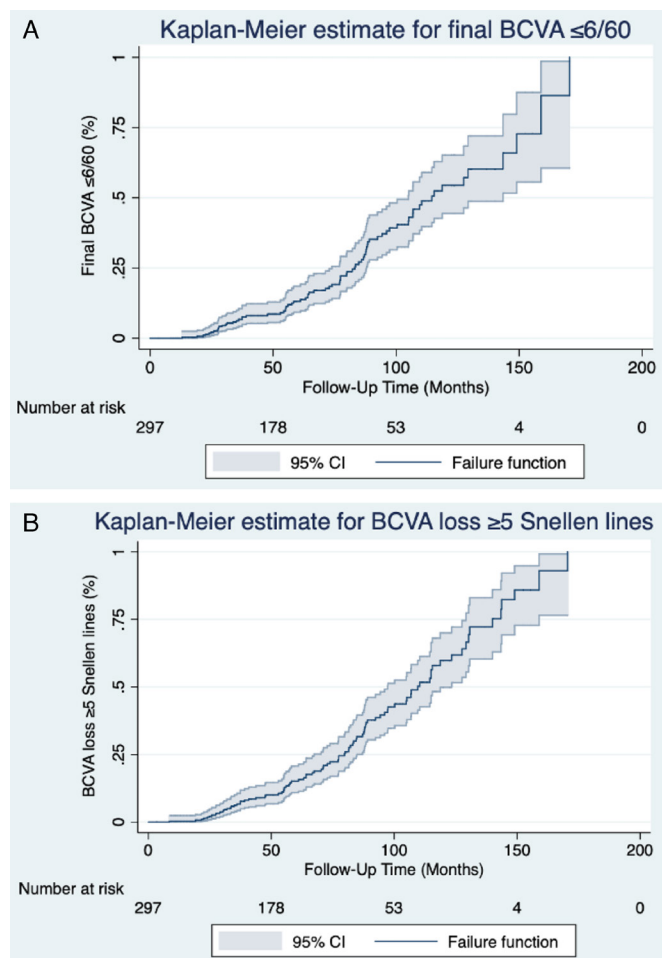


Figure 1 Kaplan-Meier estimates of visual acuity loss. (A) Final BCVA $\leq 6/60$ and (B) visual loss ≥ 5 Snellen lines. BCVA, best-corrected visual acuity.

a large series of eyes with small CMs with a median thickness of 2 mm and LBD of 7.1 mm.

Comparison with previously reported visual outcomes

Our analysis found poor visual acuity (BCVA ≥ 1.0 logMAR) 5 years after treatment only in 13.5% of cases and at 10 years in half of the cohort (54.5%). The number of eyes losing more than 5 Snellen lines was 15.1% at 5 years and 59.8% at 10 years. Vision decline occurred mostly between the 5th and the 10th year from treatment, and untreated radiation maculopathy eyes had worse final BCVA (mean 1.0 logMAR).

A previous study on Ru-106 brachytherapy in 161 cases of CM < 10 mm in LBD, of whom 60 with thickness < 2.4 mm,¹⁸ reported similar visual outcomes, with median BCVA 5 years after treatment between 20/32 and 20/30, depending on plaque size. However, 10 and 15 mm plaques were used while we used mostly 20 mm plaques (84.5% of eyes). Damato *et al* in their cohort of 458 CMs treated with Ru-106 plaque, including 106 tumours with thickness ≤ 2.5 mm, reported an estimate of BCVA $\geq 20/200$ in 92% and 57% of cases at 5 and 9 years from treatment, respectively.²¹ These figures are in keeping with our results of BCVA $\geq 20/200$ in 86.8% of cases at 5 years and 45.5% at 10 years.

More attention has been paid to I-125 brachytherapy. A large study² on choroidal and ciliary body melanomas < 3.4 mm in thickness treated with I-125 found worse visual results at 5

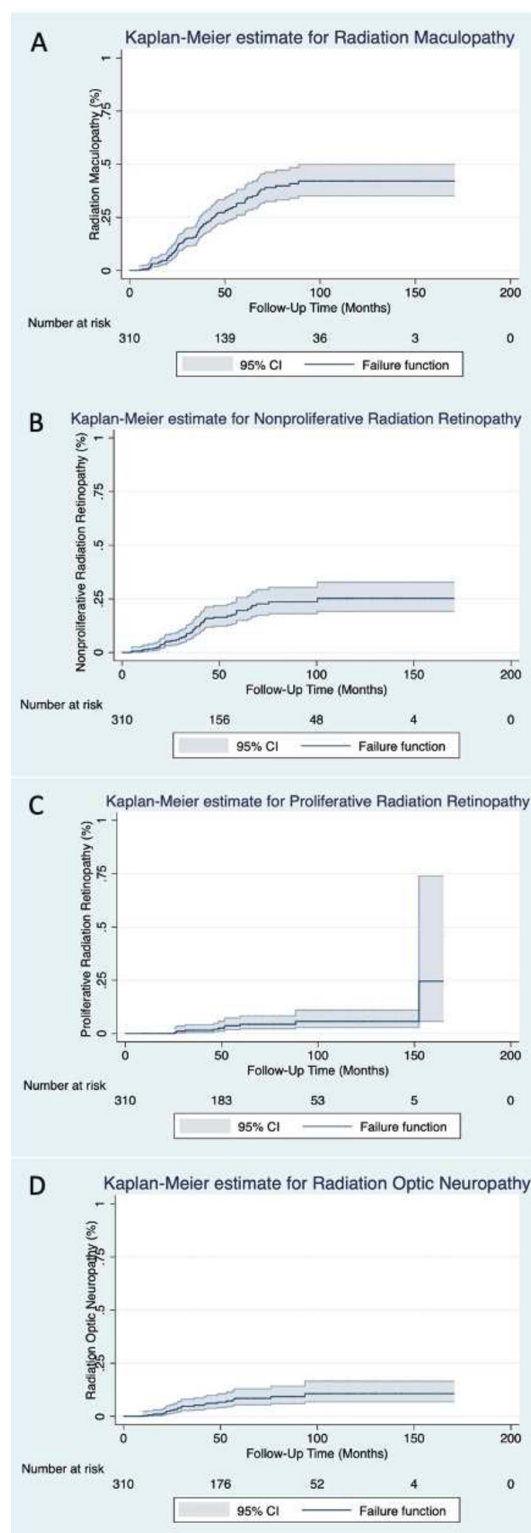


Figure 2 Kaplan-Meier estimates showing the proportion of late side effects. (A) Radiation maculopathy, (B) non-proliferative radiation retinopathy, (C) proliferative radiation retinopathy and (D) radiation optic neuropathy.

years, with a BCVA $\leq 6/60$ in 38.2% of cases (vs 13.5% in our cohort). This is due to the fact that ruthenium is a beta radiation emitter, whereas iodine is a gamma and X-ray emitter and hence has greater penetration. Interestingly, at 10 years, the visual results of the two isotopes are similar, with BCVA ≥ 1.0 logMAR

in 53.5% of cases versus 54.5% of our study. Other studies on smaller cohorts of CMs treated with I-125 report worse long-term visual outcomes, with final BCVA >1.0 logMAR in more than half of patients¹⁶ (vs 25.9%). Yupari *et al*¹⁹ and Singh *et al*²⁰ reported better visual outcomes at 3 years from treatment, as in their cohorts of 161 and 167 small CMs, a BCVA $\leq$ 0.3 logMAR was retained in 87% and 75% of eyes, respectively.

In common with the report of Shields *et al*,⁹ we found that factors with the greatest predictive value for poor visual acuity are older age and posterior tumour location, but in our study proximity to fovea, SRF and increasing tumour thickness were significant only on univariate analysis.

The only report on Palladium-103 (Pd-103)¹⁷ brachytherapy in small CM is in line with our results, with mean final BCVA of treated and untreated maculopathy patients of 0.5 and 1.0 logMAR, respectively.

Comparison with previously reported side effects

We found a 5-year probability of radiation maculopathy of 31.7%. In the only earlier report¹⁸ on Ru-106 outcomes for small CM, the incidence of radiation maculopathy at 5 years was lower at 18%–19%, but our study included also tumours with LBD >10mm and treated with larger plaques. I-125 brachytherapy is associated with higher maculopathy rates at 5 years (46.4%), as the beta-radiations of Ru-106 have a three-fold faster dose fall-off compared with gamma-radiations of I-125, implying limited depth of penetration and lower energy to surrounding structures. Similarly, Pd-103¹⁷ causes higher incidence (43.1%) and earlier onset (27 vs 32.5 months) of radiation maculopathy.

In our cohort, the 5-year estimates were 19.6% for non-proliferative and 4.4% for proliferative retinopathy, and at 10 years, 25.2% and 5.7%, respectively. The highest reported incidence of retinopathy with I-125 was 55.6% (type not specified).¹⁶ Non-proliferative retinopathy incidence for brachytherapy in small CM has not been reported before, while the reported 5-year cumulative incidence of disc neovascularisation and vitreous haemorrhage is 2.1% and 4%–12%, respectively.^{2 16 18}

For radiation optic neuropathy, we observed a 5-year risk of 8.5%, which is lower than I-125 (20.6% and 31.1%)^{2 16} and Pd-103 (20.8%),¹⁷ and similar to the previous Ru-106 report (9%),¹⁸ showing similar cumulative incidence also at 10 years (8% vs 10.7%).

While optic neuropathy and retinopathy showed similar incidence at 5 and 10 years, maculopathy incidence differed significantly (32% vs 42%) explaining the visual loss in the long term (from 13% at 5 years to 54% at 10 years).

Our study has limitations: the retrospective nature limits the quality of results; the anti-VEGF discontinuation for lack of immediate visual benefit in some patients might have affected the rate of visual deterioration and visual outcomes. It would have been interesting to explore the potential correlations between tumour pathological features such as chromosomal abnormalities and genetic mutations and clinical outcomes, however, no histopathological data were available for our sample. On the other hand, points of strength are the large dataset and the long follow-up available for the majority of patients which allowed the identification of late side effects.

Several techniques have been developed to preserve vision. Eccentric plaque placement prevents irradiation of vital structures but is technically challenging, and warrants close follow-up due to risk of edge recurrence.²² Better visual outcomes for posterior small CMs are achieved with photodynamic therapy

(PDT), but long-term follow-up studies reported local recurrence rates ranging from 20% at 3 years²³ to 54% at 5 years²⁴ and depending on several factors, particularly tumour pigmentation,²⁵ and advise its use cautiously, such as cases not amenable to other treatments. TTT was introduced and replaced argon laser for the management of small CM located in proximity of the macula and the optic disc due to a significantly lower risk of side effects.²⁶ In many ocular oncology centres, TTT was combined with plaque brachytherapy to reduce the radiation dose and achieve tumour control and vision preservation by sparing adjacent normal tissues.²⁷ Shields *et al* found that primary TTT in CM with high risk features had recurrence rates of 28% after a median of 35 months, advising alternative treatment modalities as the primary management option for tumours with multiple risk factors.²⁸ Moreover, extrascleral extension caused by nests of malignant cells infiltrating the sclera is a known complication of TTT.²⁹ The combination of PDT and TTT has also been evaluated for small CM and showed recurrence rates of 19% after a mean of 35.6 months, suggesting that close follow-up is warranted to allow timely identification and treatment of recurrences.³⁰ In our centre, TTT is not commonly used as a primary treatment but reserved for recurrences with thickness <3 mm.

A new promising therapeutic option for small CM is Bel-sar (belzupacap sarotalocan, Aura-011), a recombinant papillomavirus-like particle conjugated with a phthalocyanine photosensitizer (IRDye 700DX) activated by a near-infrared 690 nm laser light that exerts a selective cytotoxic effect on uveal melanoma cells minimising the collateral damage to adjacent structures.³¹ Any new technology will be benchmarked for tumour control and side effects against the gold standard of plaque radiotherapy.

In summary, this analysis of 310 eyes with small CM treated with Ru-106 plaque brachytherapy revealed poor visual acuity at 5 years in 13.2% and at 10 years in 54.5%, early and late side effects in 20.6% and 42.6% of eyes, respectively. The 5-year risk for maculopathy was 31.7%, for non-proliferative retinopathy was 19.6%, for proliferative retinopathy was 4.4% and for optic neuropathy was 8.5%. Radiation maculopathy was the most common side effect and eyes receiving anti-VEGF treatment showed better visual outcomes compared with untreated eyes.

The present study adds important information on the burden on visual function of globe conserving brachytherapy in small CM and will help practitioners to define the risk of side effects from plaque radiotherapy for small tumours and offer support for comparison with other treatment modalities.

Acknowledgements The research was supported by the National Institute for Health Research (NIHR) Biomedical Research Centre based at Moorfields Eye Hospital NHS Foundation Trust and UCL Institute of Ophthalmology. The views expressed are those of the author(s) and not necessarily those of the NHS, the NIHR or the Department of Health. We also acknowledge the contributions of our late colleagues, Mr John Hungerford, FRCS and Ms Victoria Cohen, FRCOphth in treating many of the patients reported herein. This manuscript was presented in part at the 46th Annual Meeting of the Macula Society, Miami, USA, and at the 106th Oxford Ophthalmological Congress, Oxford, UK.

Contributors BG and MSS contributed to the study concept and design. BG, RH, RA-J, HK, IS, GH, AKA and MSS contributed to the acquisition, analysis and interpretation of data. BG and MSS contributed to the drafting of the manuscript and critical revision for important intellectual content. BG, RH, RA-J, HK, IS, GH, AKA and MSS contributed to the final approval of the version to be published. BG and MSS are guarantors.

Funding The authors have not declared a specific grant for this research from any funding agency in the public, commercial or not-for-profit sectors.

Competing interests None declared.

Patient consent for publication Not applicable.

Ethics approval This study involves human participants. The study was approved by Moorfields Eye Hospital clinical audit department (number 53), retrospective collection of anonymised data and was carried out in compliance with the Declaration of Helsinki. All patients provided informed consent at the time of treatment.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement All data relevant to the study are included in the article or uploaded as supplementary information.

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